

CSE563 — Assignment 4

Unit IV: Graph Theory

Maps to CO4 (analyse different graph theory concepts and use them in solving practical problems).
Marks: 50. **Attempt all questions. Where an algorithm is traced, show the table of every step.**

Part A — Short Answer and Definitions (10 marks, 2 each)

Q1. [2 marks] State the handshaking lemma. Hence determine whether a graph can exist with degree sequence $(5, 5, 4, 3, 2, 1)$, and whether one can exist with $(4, 4, 4, 4, 4)$.

Q2. [2 marks] Give the condition for a connected graph to have an Eulerian circuit and for it to have an Eulerian path. Why is there no comparably simple test for a Hamiltonian cycle?

Q3. [2 marks] Prove that a tree with n vertices has exactly $n - 1$ edges.

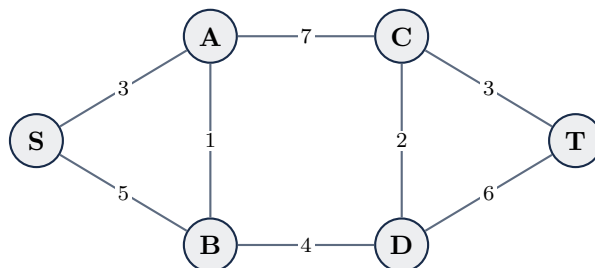
Q4. [2 marks] Explain why Dijkstra's algorithm fails on graphs with negative edge weights. Give a small counterexample.

Q5. [2 marks] Define the chromatic number. State the bounds $\omega(G) \leq \chi(G) \leq \Delta(G) + 1$ and explain in one line why each holds.

Part B — Algorithm Tracing (24 marks)

Q6. [6 marks] For the graph with adjacency lists 1:[2, 3, 4], 2:[1, 5], 3:[1, 5, 6], 4:[1, 6], 5:[2, 3, 7], 6:[3, 4, 7], 7:[5, 6]: (a) Perform BFS from vertex 1, giving the visit order and the distance of every vertex. (b) Perform DFS from vertex 1 (recursive, neighbours in the listed order), giving the visit order. (c) List the BFS tree edges and state which BFS property they demonstrate. (d) Identify one back edge found by the DFS and state what it proves.

Q7. [7 marks] Apply Dijkstra's algorithm from vertex S to the weighted graph below. Show the full table of tentative distances after each extraction, and state the shortest path and its cost to T .



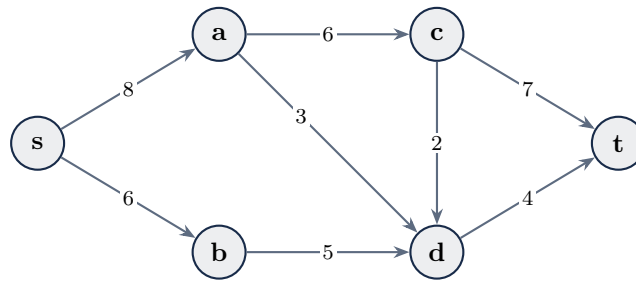
Q8. [5 marks] For the same graph as the previous question, find a minimum spanning tree using Kruskal's algorithm. List the edges in the order they are considered, mark those rejected and why, and give the total weight. Then state whether the MST equals Dijkstra's shortest-path tree, with justification.

Q9. [6 marks] Six examinations $E_1, \dots, E_6$ must be scheduled. Two exams conflict (cannot share a slot) if a student sits both. The conflicts are: E_1-E_2 , E_1-E_3 , E_1-E_4 , E_2-E_3 , E_2-E_5 , E_3-E_6 , E_4-E_5 , E_5-E_6 . (a) Draw the conflict graph and list the degree of each vertex. (b) Apply the Welsh-Powell algorithm and give the colour classes. (c) State the minimum number of slots, and prove that fewer are impossible. (d) Give the final timetable.

Part C — Network Flow (16 marks)

Q10. [10 marks] For the flow network below (source s , sink t , capacities shown): (a) Apply the

Ford–Fulkerson method, listing each augmenting path with its bottleneck and the running flow value. (b) State the maximum flow. (c) Find the minimum cut (S, T) , list the arcs crossing it, and verify its capacity equals the max flow. (d) Explain why arcs from T back into S are not counted.



Q11. [6 marks] A company must assign four engineers $\{E_1, E_2, E_3, E_4\}$ to four projects $\{P_1, P_2, P_3, P_4\}$, one each. Engineer E_1 is qualified for P_1, P_2 ; E_2 for P_2 ; E_3 for P_2, P_3 ; E_4 for P_3, P_4 . (a) Model this as a maximum bipartite matching problem and explain how to convert it into a max-flow problem. (b) Find the maximum matching and state whether a complete assignment is possible.